Response to the referees

Mode-resolved feedback gain controls spatial symmetry breaking in an explicit Notch–Delta–proneural circuit (revised as: Explicit Notch–proneural dynamics set the tempo, not the onset, of lateral-inhibition patterning)

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October 2026

I thank both referees for careful and specific reports. Each comment below is summarised in italics and followed by my response and the corresponding changes. Section, figure and equation numbers refer to the revised manuscript unless marked "v1".

Three things changed throughout:

- **Scope and claims.** The paper is reorganised around one question: which predictions require the explicit activated-Notch and proneural dynamics, and which follow from static responses alone? The Schur/static-gain identity is now presented as a standard tool, not a contribution. Material both referees regarded as known or unnecessary has been removed: the nonlinear saturation statistics, the "global rather than centred" discussion (v1 §3.3), and the stochastic check (v1 §3.4).
- **Numerics.** Every statistic was recomputed with a corrected steady-state solver and an exact-eigenvector initial state.
- **Reproducibility.** Every number has a generating script in the public repository. A test suite (341 checks) verifies each analytic statement on random parameter sets. The v1 code and data are preserved unchanged in an archive folder of the repository.

Referee 1

1.1 *Steady-state failures distort the global statistics: 733 of 4000 samples were labelled as having no positive fixed point, although the homogeneous equilibrium exists uniquely and can be found by scalar bracketing. Recalculation gives 36.325% instead of 30.3%. Solver failure must be distinguished from nonexistence.*

I agree on all points. The homogeneous steady state reduces to a scalar equation $F(D) = 0$ (Eq. 3). F is strictly decreasing, with $F(0) > 0$ and $F(\lambda_D/d_D) < 0$, so for every positive parameter set the steady state exists, is unique, and lies in $(0, \lambda_D/d_D)$ (§2.3, Supplementary Note 1). It is now computed by Brent bracketing.

- All 733 v1 "no fixed point" samples have a steady state, with residual below 3×10^{-15}.
- The nonuniform-instability fraction is **1453/4000 = 36.325%**, which reproduces the referee's value (§3.2).
- The two-parameter maps (v1 Fig. 2C) had 115 missing grid points from the same cause. They have been recomputed (Fig. 3C,D).

- Every derived statistic (classification counts, gain-criterion accuracy, dominant-mode counts) has been recomputed.
- A test now checks all 733 previously failed samples.
- The v1 statement "multistable branches found: 0" is replaced by the uniqueness proof.

1.2 *The explanation of the DC-gain misclassifications is incorrect. All seven originally reported mismatches have dominant mode $\mu = -3$; e.g. sample 1730 has gain ≈0.366 but unstable eigenvalues ≈0.159±0.262i. The failures should be explained within the tested mode, without conflating the stationary zero-eigenvalue condition with general stability.*

The referee is right, and the v1 explanation ("another mode can dominate") was wrong. The revised §3.1 and §3.3 explain the mismatches within the tested mode:

- **The plant is Hurwitz.** For every positive parameter set and every graph eigenvalue, the plant matrix A_μ is Hurwitz (§3.1(i), proved). Hence $\det(A_\mu)(-d_S) > 0$, and by Eq. 5 $\operatorname{sign}(1 - G) = \operatorname{sign}\det J_\mu$.
- **What G counts.** $G(\mu) = 1$ exactly at a zero eigenvalue, and $G(\mu) > 1$ exactly when there is an odd number of positive real eigenvalues. $G < 1$ is explicitly stated not to be a stability certificate.
- **The mismatches.** With the corrected sample there are 12, including the 7 of v1. All have dominant mode $\mu = -3$. Each has **exactly two** right-half-plane eigenvalues in the $\mu = -3$ block: a complex pair at 6 points (including 1730: $G = 0.366, 0.159 \pm 0.262i$) and two real eigenvalues at the other 6. An even count leaves $\det J_{-3} > 0$, so these instabilities are invisible at zero frequency.
- **Complete test.** The complete per-mode test is the Nyquist criterion, which is valid here because the open loop has no unstable poles. For sample 1730, $L_{-3}(i\omega)$ crosses the real axis at 1.56 when $\omega = 0.36$, and the curve encircles $+1$ twice (Fig. 4B). Encirclement counts equal the eigenvalue counts at all 4000 samples.
- **Text changes.** The v1 limitation sentence attributing the mismatches to another mode has been deleted. The stationary condition and general stability are now kept separate.

1.3 *Componentwise scaling by equilibrium concentrations changes the eigenvector and introduces mixed-mode transients. The late-window slope does match, but the initialization should be corrected or justified and the actual validation window displayed.*

Corrected. Each simulation now starts from $x^* + \varepsilon v \cos\phi$, where the whole eigenvector v is multiplied by a single scalar (Methods). The fitting window was fixed in advance: from $t = 0$ until the three-sublattice amplitude of D reaches $10^{-3}D^*$. It is drawn in Fig. 5C. With the corrected start:

point	predicted	fitted	window
P2	0.103067	0.103077	[0, 67.25]
P3	0.216300	0.216305	[0, 32]

The growth is exponential from $t = 0$, with no transient. Fig. S1 shows the v1 start for comparison. It gives early slopes of 0.443 (P2) and 0.568 (P3), the mixed-mode transient the referee identified. At P2 it also produces a long intermediate plateau before the final pattern is reached. The v1 phrases "frozen validation window" and "the earliest fit contains initial transient contamination" are removed.

1.4 *The unit-DC-gain/zero-eigenvalue equivalence follows from the Schur-complement identity under the relevant nonsingularity conditions; numerical agreement verifies consistency but does not establish novelty. A substantive comparison with graph-based approaches and a matched reduced model is needed to identify predictions that specifically require the explicit downstream dynamics.*

I agree. The identity is now presented as standard (§1, §3.1, §4). It is related explicitly to graph-based static-gain criteria [11, 12] through $\det J_\mu = \det J_{\mathrm{loc}} \det(I - \mu K)$, where K is the static input–output gain of an isolated cell (Methods; verified to 10^{-15}). The revision adds the two comparisons requested (§3.4, Fig. 5, Table 2).

Matched reduced models. Proneural activity, or both activated Notch and proneural activity, are slaved by quasi-steady-state elimination (DNA and DN reductions). The DNA reduction is a Sprinzak-type three-component Notch–Delta–reporter model. Both reductions have exactly the same steady states, homogeneous and patterned, and their Jacobians are Schur complements of J_μ. They therefore share the stationary boundary, the critical λ_N, and the selected mode exactly. These predictions do **not** require the explicit dynamics, and the manuscript now says so.

Predictions that do require the explicit dynamics:

- **Growth rate.** Along a proneural-timescale family that leaves every steady state and every static gain fixed, $\lambda_N^c = 1.304815343$ is unchanged for $\alpha \in [10^{-2}, 10^{4}]$. The P2 growth rate varies from 8×10^{-4} to 1.59. When proneural turnover is slow, the rate is $\approx d_S[G-1]$. Over the global sample, the matched DNA reduction overestimates the growth rate at every unstable point (median 4.9-fold; 15.5-fold at P2). Nonlinear simulations confirm this (pattern at $t \approx 111$ versus ≈ 7), and both models reach the same final pattern, to within 7×10^{-11}.
- **Non-stationary instabilities.** The DNA reduction predicts 26 instabilities where the explicit model is stable (DN reduction: 47), all with $G < 1$. In the other direction, the static criterion misses the 12 two-eigenvalue instabilities of point 1.2.

A further analytic result has been added (§3.1(iii)): the static gain is maximised at the most negative graph eigenvalue for **all** positive parameters, despite the two-channel trans coupling and cis inactivation. The stationarily selected mode is therefore the three-sublattice mode everywhere. This also answers Referee 2's point 2.3.

Referee 2

2.1 *Variables are used before being defined (e.g. in the abstract and on p. 1, l. 57, not defined until p. 5); (m, n) is undefined on p. 5, l. 45. The paper needs serious editing.*

The manuscript has been rewritten with a definition-before-use rule:

- The abstract contains no symbols.
- The model, every variable and parameter (also tabulated in Table 1), the graph eigenvalue μ, and the plane-wave indices (m, n) with their wavevectors are all defined in §2, before any result.
- Fig. 1D now shows $\mu(m, n)$ over all indices.
- Symbols in captions are defined within each caption.

2.2 *The novelty relative to previous Notch–Delta work is unclear; checking full-model stability from the exact Jacobian under a pattern assumption is not a contribution.*

I agree that the per-mode Jacobian check is not a contribution. It has been reduced to one sentence in Methods. The revised paper states its contributions in §4 ("What is and is not new"):

- uniqueness of the steady state for all positive parameters;
- a proof that cis inactivation and two-channel trans coupling never change the stationarily selected mode;

- an exact separation of predictions that depend only on static responses, which every matched reduced model shares, from those that depend on downstream timescales, which reduced models get wrong;
- a quantitative account of where a static stability test fails.

2.3 *There could be cases in which a pattern other than the assumed one is unstable, but no such case is presented.*

This is now addressed directly (§3.1(iii) and §3.3).

- **Proof for stationary instabilities.** For stationary instabilities no other pattern can come first. $G(\mu) < G(\mu_{\min})$ for every other graph eigenvalue, for every positive parameter set and every regular contact graph.
- **Search for non-stationary cases.** The proof does not cover non-stationary instabilities, so 40,000 additional random parameter sets were searched, half in a box ten times wider than the original. Of these, 13,047 were unstable. None had a nonuniform mode other than $\mu = -3$ dominant, or unstable while $\mu = -3$ was stable.
- **The one other unstable mode.** The only other unstable mode found is the uniform mode, through a complex pair (synchronous oscillation), at 45 of the 4000 sampled points. It always coexists with an unstable three-sublattice mode, and it is reported as such.
- **Mismatches within $\mu = -3$.** The twelve static-gain mismatches (point 1.2) are a different kind of case. The tested mode itself is unstable through two eigenvalues.

2.4 *That the system can be stable to uniform perturbations while unstable to patterned ones is the same as in all other Notch–Delta models.*

Agreed. This is no longer presented as a finding. It appears once, as a one-line consequence of §3.1(iii) ($G(6) < 0$, so the uniform mode cannot lose stability through a zero eigenvalue). In §4 it is explicitly described as a generic feature of lateral-inhibition models.

2.5 *Putting equations as subfigures is not good practice.*

Removed. The v1 Fig. 1D (equation box) and Fig. 3A–B (block diagram with equations and the Schur identity) no longer exist. All equations are in the text and are numbered.

2.6 *Belabouring the growth of the linear instability into a stable nonlinear pattern is unnecessary given earlier work.*

Agreed. The v1 Fig. 5 is removed: amplitude order parameter, snapshots, Delta histogram, solver comparison and Moran's I. Nonlinear simulations now appear only where they test a new claim (Fig. 5C,D): that the explicit model and its matched reduction grow at different, predicted rates yet reach the same final pattern. The paragraph on fate commitment is cut to one sentence in the Limitations.

2.7 *Sections 3.1–3.2 (distinction from simpler three-component models) are the original part; §3.3 seems trivial; the results of §3.4 are not shown in figures and their relevance to stochasticity is unclear.*

Thank you. The revision is built around the part the referee found original. The comparison with three-component models is now quantitative: an explicitly constructed matched three-component (DNA) reduction, analysed in §3.4 and Fig. 5. v1 §3.3 has been deleted. v1 §3.4 (the stochastic check) has been removed from the manuscript, since it was not needed for any conclusion. Its code remains in the repository archive.

Summary of changes

v1	revised
30.3% unstable (733 solver failures excluded)	36.3% unstable; steady state proved unique
mismatches attributed to “another mode”	two RHP eigenvalues in the $\mu = -3$ block; Nyquist count
componentwise-scaled initial state, “frozen” window	exact eigenvector; window fixed in advance and drawn
unity-gain identity presented as a key result	presented as standard; equivalence to graph static-gain criteria
no reduced-model comparison	matched DNA and DN reductions; timescale family; Table 2
equations inside Figs. 1D, 3A–B	equations only in text
nonlinear saturation figure (v1 Fig. 5)	removed; one targeted comparison kept
v1 §3.3 and §3.4 (stochastic)	removed
—	proof that stationary onset always selects the three-sublattice mode